CSCI E-106:Assignment 2 - Solutions

Preparations (Libraries, basic functions)

Due Date: February 13, 2026 at 11:59 pm EST

Instructions

Students should submit their reports on Canvas. The report needs to clearly state what question is being solved, step-by-step walk-through solutions, and final answers clearly indicated. Please solve by hand where appropriate.

Please submit two files: (1) a R Markdown file (.Rmd extension) and (2) a PDF document, word, or html generated using knitr for the .Rmd file submitted in (1) where appropriate. Please, use RStudio Cloud for your solutions.

Problem 1

Refer to the CDI data set. Use the number of active physicians as Y and total personal income as X. (50 points) ## data preparation-Check data, rename variables and select only the variables you plan to use to make your life easier

```r
PMQ1<- read.csv("CDI Data.csv")

summary(PMQ1)
```

```
##  Identification.number    County              State             Land.area
##  Min.   :  1.0          Length:440         Length:440         Min.   :   15.0
##  1st Qu.:110.8          Class :character   Class :character   1st Qu.:  451.2
##  Median :220.5          Mode  :character   Mode  :character   Median :  656.5
##  Mean   :220.5                                                Mean   : 1041.4
##  3rd Qu.:330.2                                                3rd Qu.:  946.8
##  Max.   :440.0                                                Max.   :20062.0
##  Total.population  Percent.of.population.aged.18.34
##  Min.   : 100043   Min.   :16.40
##  1st Qu.: 139027   1st Qu.:26.20
##  Median : 217280   Median :28.10
##  Mean   : 393011   Mean   :28.57
##  3rd Qu.: 436064   3rd Qu.:30.02
##  Max.   :8863164   Max.   :49.70
##  Percent.of.population.65.or.older Number.of.active.physicians
##  Min.   : 3.000                    Min.   :   39.0
##  1st Qu.: 9.875                    1st Qu.:  182.8
##  Median :11.750                    Median :  401.0
##  Mean   :12.170                    Mean   :  988.0
##  3rd Qu.:13.625                    3rd Qu.: 1036.0
##  Max.   :33.800                    Max.   :23677.0
##  Number.of.hospital.beds Total.serious.crimes Percent.high.school.graduates
##  Min.   :   92.0         Min.   :   563       Min.   :46.60
```

```
##  1st Qu.:  390.8        1st Qu.:  6220       1st Qu.:73.88
##  Median :  755.0        Median : 11820       Median :77.70
##  Mean   : 1458.6        Mean   : 27112       Mean   :77.56
##  3rd Qu.: 1575.8        3rd Qu.: 26280       3rd Qu.:82.40
##  Max.   :27700.0        Max.   :688936       Max.   :92.90
##  Percent.bachelor.s.degrees Percent.below.poverty.level Percent.unemployment
##  Min.   : 8.10              Min.   : 1.400              Min.   : 2.200
##  1st Qu.:15.28              1st Qu.: 5.300              1st Qu.: 5.100
##  Median :19.70              Median : 7.900              Median : 6.200
##  Mean   :21.08              Mean   : 8.721              Mean   : 6.597
##  3rd Qu.:25.32              3rd Qu.:10.900              3rd Qu.: 7.500
##  Max.   :52.30              Max.   :36.300              Max.   :21.300
##  Per.capita.income Total.personal.income Geographic.region
##  Min.   : 8899     Min.   :  1141        Min.   :1.000
##  1st Qu.:16118     1st Qu.:  2311        1st Qu.:2.000
##  Median :17759     Median :  3857        Median :3.000
##  Mean   :18561     Mean   :  7869        Mean   :2.461
##  3rd Qu.:20270     3rd Qu.:  8654        3rd Qu.:3.000
##  Max.   :37541     Max.   :184230        Max.   :4.000
```

```r
Y <- PMQ1$Number.of.active.physicians
X <- PMQ1$Total.personal.income

CDI_data<-data.frame(cbind(Y,X))
```

#a-) Built a regression model to predict Y by using X. Write down the stand error of the slope and intercept. (5 points)

CONCLUSION Using the R-lm() function, we built "Y = -48.394849 + 0.131701X", we then extracted the values of the slope and the intercept estimates and their corresponding standard errors from the summary(model) object and then we illustrated the results in the Table below.

```r
model<-lm(Y ~ X, data = CDI_data)
Estimate_b0<-coef(summary(model))[1, "Estimate"]
Estimate_b1<-coef(summary(model))[2, "Estimate"]

StandardErr_b0<-coef(summary(model))[1, "Std. Error"]
StandardErr_b1<-coef(summary(model))[2, "Std. Error"]

print(paste("Estimate b0: ",Estimate_b0))
```

```
## [1] "Estimate b0:  -48.3948489196034"
```

```r
print(paste("Estimate b1: ",Estimate_b1))
```

```
## [1] "Estimate b1:  0.131701189183656"
```

```r
print(paste("Standard error b0: ",StandardErr_b0))
```

```
## [1] "Standard error b0:  31.8333281771254"
```

```r
print(paste("Standard error b1: ",StandardErr_b1))
```

```
## [1] "Standard error b1:  0.0021102791021475"
```

```r
# Print the equation (refer to lines above)
dispRegFunc(model)
```

```
## [1] "Y = -48.394849 + 0.131701X"
```

#b-) Are the slope and intercept statistically significant? Write down your null and alternative hypothesis and p value of the test. Calculate the 95% Confidence interval for the slope. (20 points)

CONCLUSION Using an alpha = 0.05, Intercept is not significant but the slope is significant. To do this test, one may apply the six steps below and we calculated the confidence interval for the slope: 2.5 % 97.5 % 0.1275537 0.1358487 using the code below:

Step 1: Write down your null and alternative hypothesis H_0: Regression parameters(i) is equal to zero H_a: Regression parameters(i) is not equal to zero

Step 2: Identify the appropriate tests statistics.

Step 3: Specify the level of significance: if not given in the problem use an alpha=0.05

Step 4: Decision Rule for the test: We reject the null hypothesis if the calculated t-statistic is larger than the absolute value of the t critical value.

Step 5: Calculate the test statistic.

Step 6: Make the Decision.

Parameter	Estimate	Std.Error	t statistic	t critical	p-value	Decision
Intercept	-48.394849	31.83332818	-1.52026	1.965	0.12917	Accept H0
Slope	0.131701	0.002110279	62.40937	1.965	0.00000	Reject H0

#c-) Predict the number of active physicians for total personal income of 100000,250000 and 50000. Calculate the 95% confidence interval (Hint: Use the range of total personel income to identify which observations are within the range of the training data) (20 points)

CONCLUSION Using an alpha = 0.05, the Doctors given the incomes within the range and corresponding confidence intervals are: fit lwr upr 6,536.665 6353.955 6719.374 fit lwr upr 13,121.72 12735.9 13507.55 And for the income outside the range fit lwr upr 32,876.9 31371.92 34381.89 See code below

```
options("scipen" = 12)
# Step 1: Find the range of the personal income to see if 100000,250000 and 50000 are included.
range_values<-range(X)
maximum <-max(X)
# Step 2: 95% CI for the estimated mean within range
# 100000 and 50000 are in range, so when predicting we are interpolating within development data
ci<-predict(model, data.frame(X=50000), interval = "confidence", level=0.95)
ci
```

```
##        fit      lwr      upr
## 1 6536.665 6353.955 6719.374
```

```
ci <- predict(model, data.frame(X=100000),interval = "confidence", level = 0.95)
ci
```

```
##        fit     lwr      upr
## 1 13121.72 12735.9 13507.55
```

```
# Step 3: 95% CI for the predicted mean given the new observation out of range
# The Large Value 250,000 is out of range, we use prediction interval to account for uncertainty
ci <- predict(model, data.frame(X=250000), interval = "prediction", level = 0.95)
ci
```

```
##       fit      lwr      upr
## 1 32876.9 31371.92 34381.89
```

#d-) What is the error variance of the regression line?(5 points)

CONCLUSIONThe residual variance of the model is 324539.4, see code below.

```
variance<-(summary(model)$sigma)^2
variance
```

```
## [1] 324539.4
```

```
# OR
Error_variance = sum(residuals(model)^2) / df.residual(model)
```

Problem 2

Refer to the CDI data set. Use the number of active physicians as Y and total personal income as X. Select 1,000 random samples of 400 observations, fit the regression model and record β_0 and β_1 for each selected sample. Calculate the mean and variance of β_0 and β_1 based on the 1,000 different regression line and compare against the regression model in question 1 part b. (50 points)

CONCLUSIONsee code below. We use the R-function sample() to generate random row indices (of size 400) and subset the original data frame (size is 440) without replacement. To compare against the original regression model we averaged the results. The parameter estimates are pretty close, however the simulated Standard Error of the intercept and slope were lower.

```
set.seed(123)
# Data frame with columns X and Y
prg.hw2 <- function(df, n_sim, sample_size) {
  # Initialize matrix for b0 and b1
  out <- matrix(0, nrow = n_sim, ncol = 2)
  colnames(out) <- c("beta0", "beta1")

  for (i in 1:n_sim) {
    # 1. Randomly select row indices
    indices <- sample(1:nrow(df), size = sample_size, replace = FALSE)

    # 2. Subset the data
    sample_data <- df[indices, ]

    # 3. Fit regression model Y ~ X
    f <- lm(Y ~ X, data = sample_data)

    # 4. Record coefficients
    out[i, ] <- f$coefficients
  }
  return(out)
}

# Execute for 1,000 samples of 400 observations
results <- prg.hw2(df = CDI_data, n_sim = 1000, sample_size = 400)
### Calculate the mean and variance of $\beta_0$ and $\beta_1$ based on the 1,000 different regression 

head(results)
```

```
##          beta0     beta1
## [1,] -63.56131 0.1334466
## [2,] -31.89056 0.1314999
## [3,] -52.42159 0.1314951
```

```
## [4,] -67.28268 0.1319287
## [5,] -44.12390 0.1315070
## [6,] -50.70925 0.1315168
```

```r
#print for comparison between simulated average values and 1b
print(paste("Estimate b0: ",Estimate_b0))
```

```
## [1] "Estimate b0:  -48.3948489196034"
```

```r
print(paste("Estimate b1: ",Estimate_b1))
```

```
## [1] "Estimate b1:  0.131701189183656"
```

```r
print(paste("Standard error b0: ",StandardErr_b0))
```

```
## [1] "Standard error b0:  31.8333281771254"
```

```r
print(paste("Standard error b1: ",StandardErr_b1))
```

```
## [1] "Standard error b1:  0.0021102791021475"
```

```r
################################### To Compare

average_parameter<-apply(results,2,mean)
print(paste("Avg Estimate b0: ",average_parameter[1]))
```

```
## [1] "Avg Estimate b0:  -48.9238113379518"
```

```r
print(paste("Avg Estimate b1: ",average_parameter[2]))
```

```
## [1] "Avg Estimate b1:  0.131761050145195"
```

```r
average_std<-sqrt(apply(results,2,var))
print(paste("Avg StdardErr b0:",average_std[1]))
```

```
## [1] "Avg StdardErr b0: 8.38130248848631"
```

```r
print(paste("Avg StdardErr b1:",average_std[2]))
```

```
## [1] "Avg StdardErr b1: 0.00108932728021097"
```

```r
################################### To Compare
```

“‘